

Mice DEA March 2026

Sample Preparation and RNA Extraction/Sequencing (NRC)

- Acute respiratory distress syndrome (ARDS) in mouse lung tissue
- Lipopolysaccharide (LPS)-induced lung injury mouse model tissues received on 23 August, 2023
- RNA was extracted from mouse lung tissue using a Maxwell RSC SimplyRNA kit
- Libraries were prepared using the Illumina Stranded RNA kit which uses PolyA enrichment to remove rRNA
- Sequencing was performed on an Illumina NextSeq 550Dx platform
- Total of 12 biological samples (4 replicates x CFA, CFB, control)
- Each sample underwent paired-end sequencing and generated two files: R1 (forward read) and R2 (backward read) for a total of 60 FASTQ files

Count Generation (Gurkamal)

These files were received from gurkamal@canurta.com on March 22nd in a private Teams chat.

Step	Tool	Input	Output	Notes
QC	FastQC + MultiQC	Raw reads (12 R1 files)	Assessments of read quality, adapter content, duplication, and GC distributions multiqc_report.html	
Trimming	fastp	Raw reads (12 FASTQ files)	Trimmed reads (12 FASTQ files)	
Alignment	STAR	Raw reads (12 FASTQ files) Reference genome	12 SAM/BAM files	Downloaded the mm9 reference genome and the complementary ensemble GTF (annotation).
Quantification	featureCounts	12 BAM files	featureCounts_raw.txt featureCounts_raw.txt.summary	
Parsing	Pandas	featureCounts_raw.txt	gene_counts_matrix.tsv (genes x samples)	Removed the extra columns from the featureCounts table and renamed the sample columns from BAM file paths to sample IDs so the count matrix could be used in DESeq2 (genes x samples). This needs to be transposed (samples x genes) for pydeseq2.

Sample-Level Quality Control

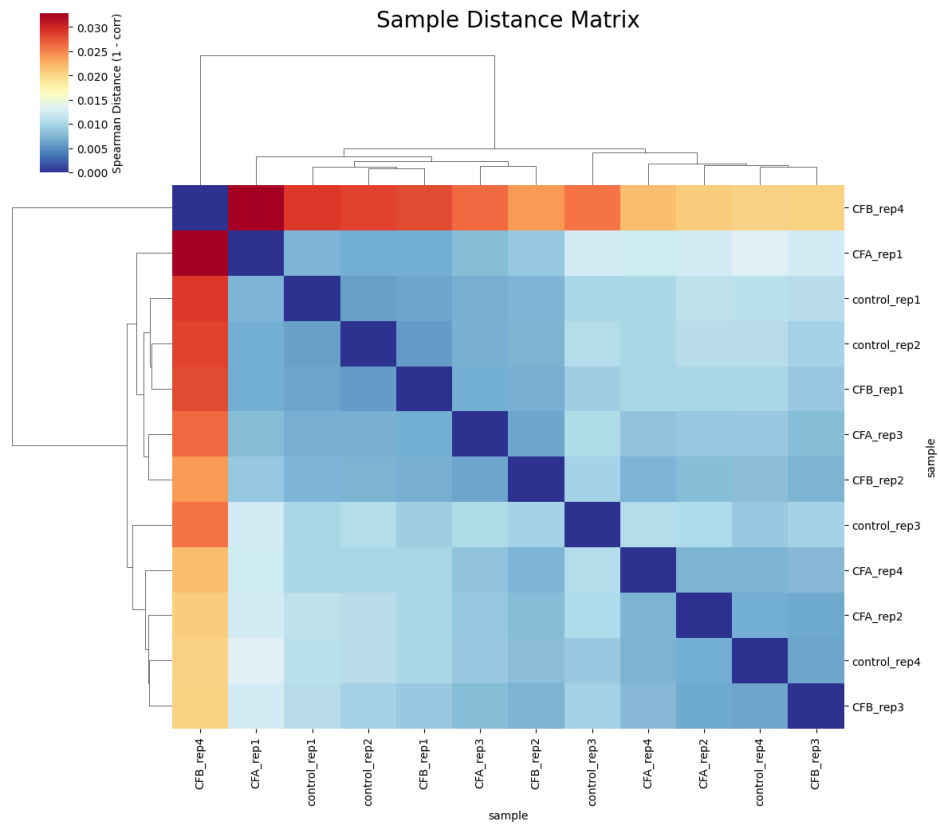
Sample Screening

Every single sample exceeded the absolute threshold for multimapping rate (0.15), indicating a global rRNA contamination or other technical error. However, the consistent signal is a positive. CFA_rep1 was the only outlier, flagged for an abnormally low mapping rate (0.66), still above the absolute threshold of 0.5.

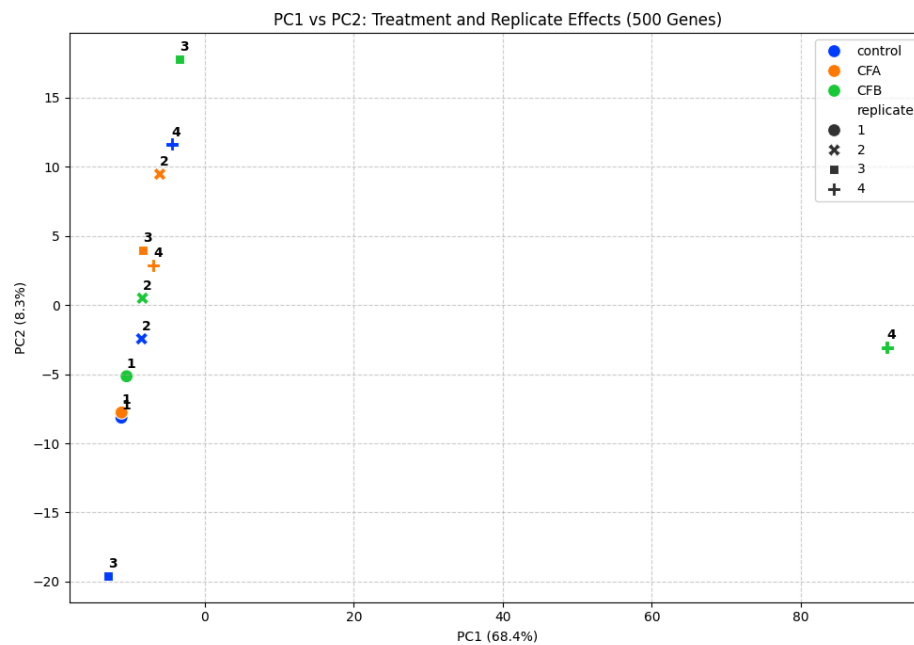
sample	relative_thresholds	total_reads	library_size	mapping_rate	n_detected_genes	multimapping_rate	absol
CFA_rep1	low_mapping_rate	34099257	22382159	0.656383	19898	0.211304	high_

Sample Outlier Detection

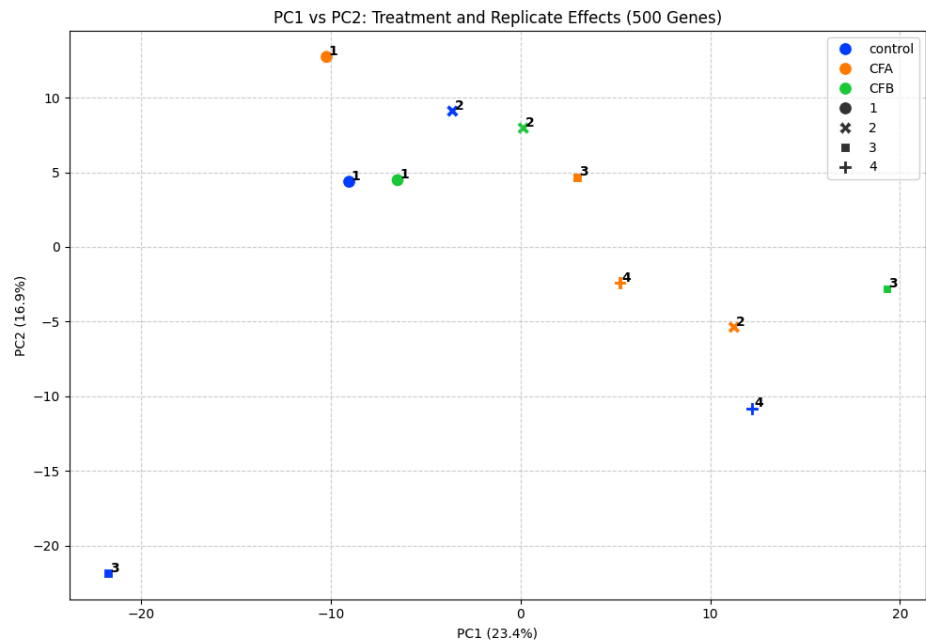
A sample correlation heatmap of all samples shows chaotic clustering, with CFB_rep4 standing out as a clear outlier.



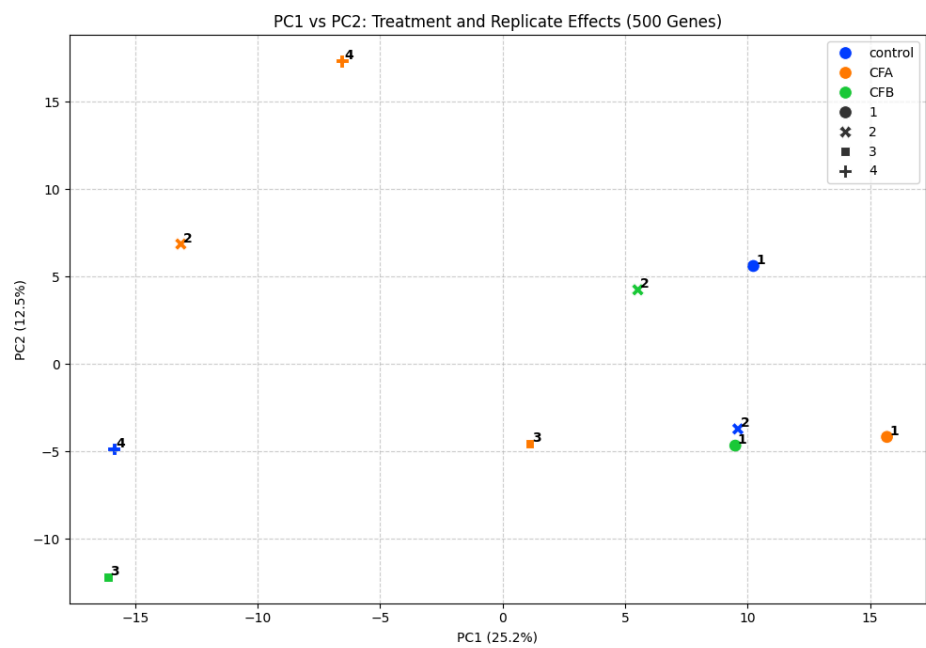
CFB_rep4 is dominating the PCA, standing out as PC1 representing 68.4% of the variance. It must be removed. It also appears that control_rep 3 may be problematic.

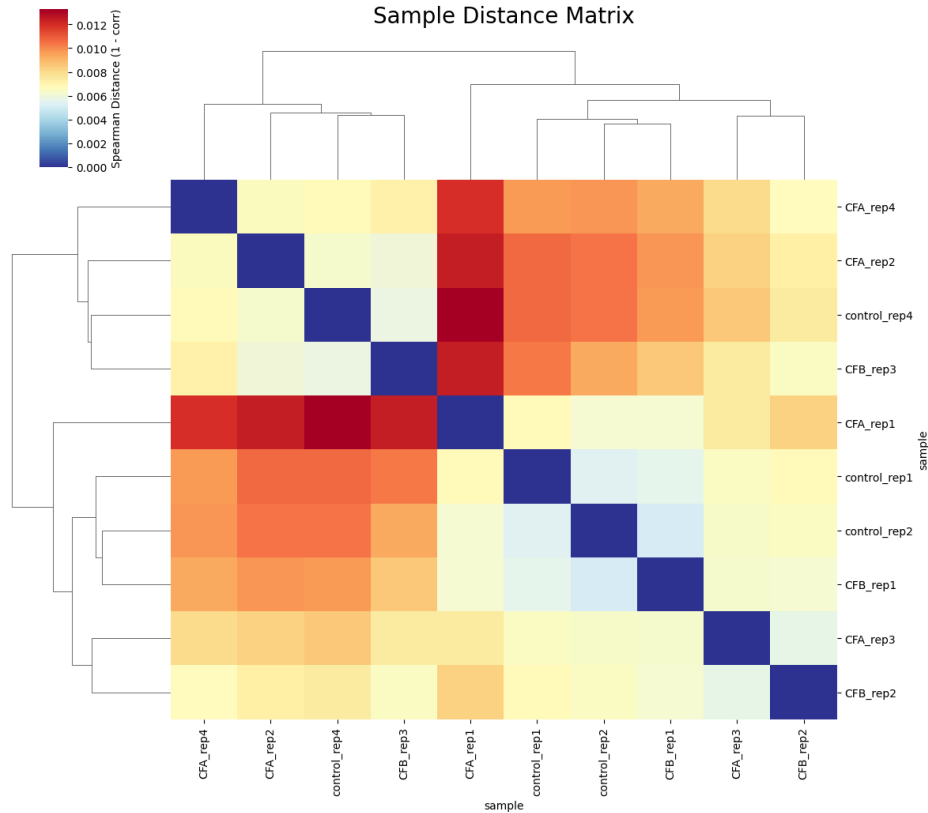


After removing CFB_rep4, PC1 (23.4%) and PC2 (16.9%) represent another outlier: control_rep3.



After removing both CFB_rep4 and control_rep3, the PCA and heatmap still look messy.





Differential Expression Analysis

Gene and Sample Filtering

10 samples entered the statistical analysis after filtering out the two outliers above. After filtering out genes with less than 10 counts, 18,902 of a total 37,992 were retained.

Metadata dataframe that entered the `DeseqDataSet`.

sample	sample_ID	treatment	replicate
control_rep1	H23-142	control	1
control_rep2	H23-143	control	2
control_rep4	H23-145	control	4
CFA_rep1	H23-146	CFA	1
CFA_rep2	H23-147	CFA	2
CFA_rep3	H23-148	CFA	3
CFA_rep4	H23-149	CFA	4
CFB_rep1	H23-150	CFB	1
CFB_rep2	H23-151	CFB	2
CFB_rep3	H23-152	CFB	3

Statistical Design

- A single `DeseqDataSet` was fit, so that there was one global GLM per gene for all 10 samples. This allows for more statistical power when estimating gene-wide dispersions since more replicates are available, helping with controlling the noise in this messy dataset.

- A single-factor (treatment) reference-means model was used with control as the reference level (intercept; β_0). The other two treatments were each given a coefficient referenced to this intercept (β_2, β_3). This setup aligns with the *Drosophila* analysis where results were in the form of treatment tested against a pathological control as reference.

The design matrix used in this analysis.

sample	Intercept (β_0)	treatment[T.CFA] (β_1)	treatment[T.CFB] (β_2)
control_rep1	1.0	0	0
control_rep2	1.0	0	0
control_rep4	1.0	0	0
CFA_rep1	1.0	1	0
CFA_rep2	1.0	1	0
CFA_rep3	1.0	1	0
CFA_rep4	1.0	1	0
CFB_rep1	1.0	0	1
CFB_rep2	1.0	0	1
CFB_rep3	1.0	0	1

The contrasts used in this analysis, of format ['variable_of_interest', 'tested_level', 'ref_level'].

```
["treatment", "CFA", "control"]
["treatment", "CFB", "control"]
```